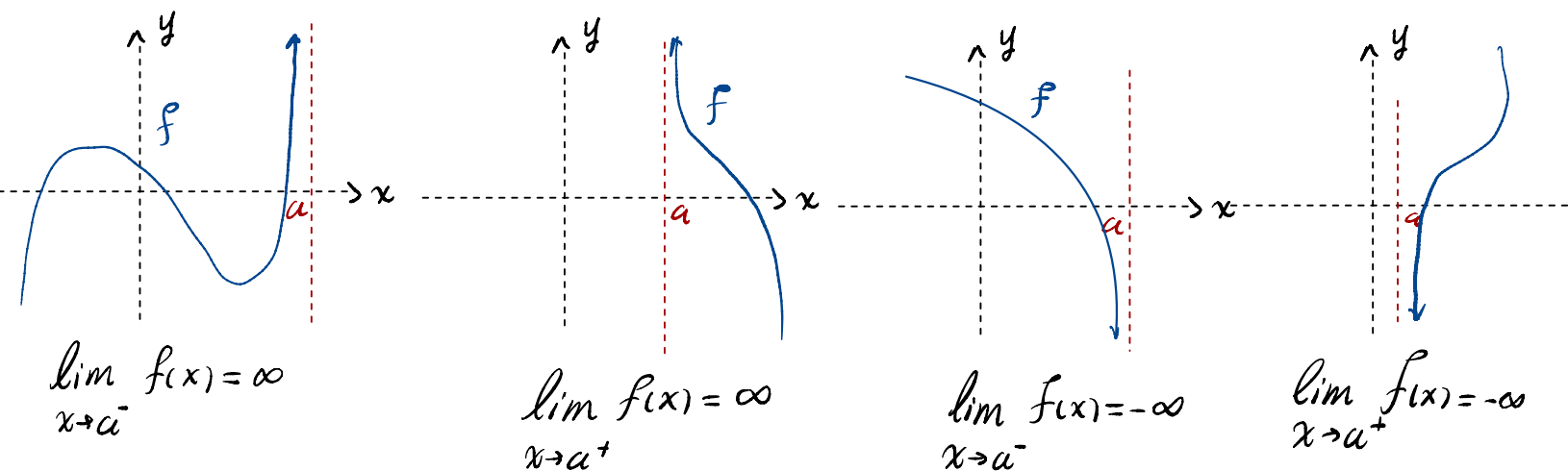


Lecture 3, Dec 19th, 2025



Def. The vertical line $x=a$ is called a vertical asymptote of the curve $y=f(x)$ if at least one of the following statements is true.

$$\lim_{x \rightarrow a} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a} f(x) = -\infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

Example. Find the vertical asymptotes of $f(x) = \tan x$.

Soln. We know that

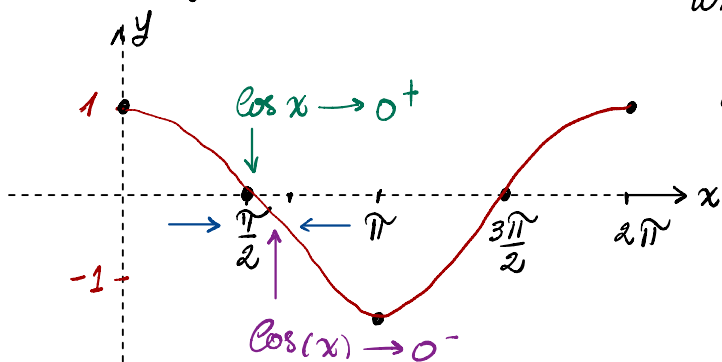
$$\tan x = \frac{\sin x}{\cos x}$$

there are potential vertical asymptotes where $\cos x = 0$ for example if $x = \frac{\pi}{2}$.

when $x \rightarrow \frac{\pi}{2}^-$ then $\cos x \rightarrow 0^+$

and when $x \rightarrow \frac{\pi}{2}^+$ then $\cos x \rightarrow 0^-$

So, there is a vertical asymptote at $x = \frac{\pi}{2}$.



In general we know that $\cos x = 0$ for $x = \frac{\pi}{2} + n\pi$ where n is an integer, and all are vertical asymptotes of $f(x) = \tan x$.

Example. Find $\lim_{x \rightarrow 0} \sin \frac{\pi}{x}$.

Sln. Notice that if we try approximate to 0 we get.

$$f(1) = \sin \pi = 0$$

$$f\left(\frac{1}{2}\right) = \sin 2\pi = 0$$

$$f\left(\frac{1}{3}\right) = \sin 3\pi = 0$$

$$f(0.1) = \sin 10\pi = 0.$$

$$f(0.01) = \sin 100\pi = 0$$

We can guess that the limit is 0, but this time our guess is wrong. Notice that if $x = \frac{1}{n}$ then $f\left(\frac{1}{n}\right) = \sin n\pi = 0$ for $n \in \mathbb{Z}$. but if $x = \frac{2}{5}$ then $f\left(\frac{2}{5}\right) = \sin\left(5\left(\frac{\pi}{2}\right)\right) = 1$.

$$x = \frac{2}{101} \text{ then } f\left(\frac{2}{101}\right) = \sin\left(101\left(\frac{\pi}{2}\right)\right) = 1$$

$$\vdots$$
$$x = \frac{2}{\text{odd \#}} \text{ then } f\left(\frac{2}{\text{odd \#}}\right) = 1$$

then $\lim_{x \rightarrow 0} \sin \frac{\pi}{x}$ does not exist.

Calculating Limits using the Limits Laws.

We use the following properties of limits, called the limits laws to calculate limits.

Thm (Limit Laws). Suppose that c is a constant and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exists. then,

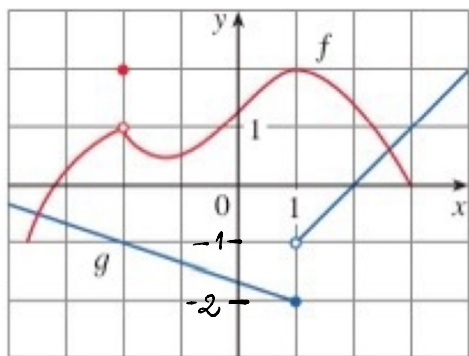
$$i). \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x).$$

$$ii). \lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot \lim_{x \rightarrow a} f(x)$$

$$\text{iii). } \lim_{x \rightarrow a} [f(x) \cdot g(x)] = \left[\lim_{x \rightarrow a} f(x) \right] \cdot \left[\lim_{x \rightarrow a} g(x) \right].$$

$$\text{iv). } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0.$$

Example.



Compute.

$$\text{a). } \lim_{x \rightarrow -2} [f(x) + 5g(x)].$$

$$\text{b). } \lim_{x \rightarrow 1} [f(x) \cdot g(x)].$$

$$\text{c). } \lim_{x \rightarrow 2} \frac{f(x)}{g(x)}. \quad \text{for you!}$$

Sol. a). $\lim_{x \rightarrow -2} f(x) = 1$

$$\lim_{x \rightarrow -2} g(x) = -1.$$

then we can apply the Law 1 and Law 2.

$$\lim_{x \rightarrow -2} [f(x) + 5g(x)] = \lim_{x \rightarrow -2} f(x) + \lim_{x \rightarrow -2} (5g(x)) \quad \text{by Law 1.}$$

$$= \lim_{x \rightarrow -2} f(x) + 5 \lim_{x \rightarrow -2} g(x) \quad \text{by Law 2.}$$

$$\underbrace{\quad}_{1} \quad \underbrace{\quad}_{-1}$$

$$= 1 + 5(-1) = -4.$$

$$\text{b). } \lim_{x \rightarrow 1} f(x) = 2$$

$$\lim_{x \rightarrow 1} g(x) = \text{DNE.}$$

$$\lim_{x \rightarrow 1^-} f(x) = 2$$

$$\lim_{x \rightarrow 1^-} g(x) = -2$$

$$\lim_{x \rightarrow 1^+} f(x) = 2$$

$$\lim_{x \rightarrow 1^+} g(x) = -1.$$

$$\lim_{x \rightarrow 1^-} [f(x) \cdot g(x)] = \lim_{x \rightarrow 1^-} f(x) \cdot \lim_{x \rightarrow 1^-} g(x) = 2(-2) = -4.$$

$$\lim_{x \rightarrow 1^+} [f(x) \cdot g(x)] = \lim_{x \rightarrow 1^+} f(x) \cdot \lim_{x \rightarrow 1^+} g(x) = 2(-1) = -2.$$

Since $\lim_{x \rightarrow 1^-} [f(x) \cdot g(x)] \neq \lim_{x \rightarrow 1^+} [f(x) \cdot g(x)]$ then

$$\lim_{x \rightarrow 1} [f(x) \cdot g(x)] = \text{DNE}.$$

If we use the product rule repeatedly we obtain the following law.

Power law.

$$\lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n \quad \text{where } n \in \mathbb{N}.$$

Root Law.

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} \quad \text{where } n \in \mathbb{N}.$$

(if n is even, we assume that $\lim_{x \rightarrow a} f(x) \geq 0$.)

- $\lim_{x \rightarrow a} c = c$
- $\lim_{x \rightarrow a} x = a$
- $\lim_{x \rightarrow a} x^n = a^n, a \in \mathbb{N}.$
- $\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}, n \in \mathbb{N}$ if n is even, $a \geq 0$.

Example. Evaluate the limits and justify each step.

a). $\lim_{x \rightarrow 5} (2x^2 - 3x + 4)$

b). $\lim_{x \rightarrow 2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$

Sln. a). $\lim_{x \rightarrow 5} (2x^2 - 3x + 4) = \lim_{x \rightarrow 5} (2x^2) - \lim_{x \rightarrow 5} (3x) + \lim_{x \rightarrow 5} 4$ by law 1.

$$= 2 \lim_{x \rightarrow 5} x^2 - 3 \lim_{x \rightarrow 5} x + \lim_{x \rightarrow 5} 4 \text{ by law 2.}$$

$$\underbrace{\lim_{x \rightarrow a} x^2 = a^2}_{x \rightarrow a} \quad \underbrace{\lim_{x \rightarrow a} x = a}_{x \rightarrow a} \quad \underbrace{\lim_{x \rightarrow a} c = c}_{x \rightarrow a}$$

$$= 2(5)^2 - 3(5) + 4 = 39.$$

b). • $\lim_{x \rightarrow 2} (5 - 3x) = \lim_{x \rightarrow 2} 5 - 3 \lim_{x \rightarrow 2} x = 5 - 3(2) = -1 \neq 0.$

Using the Law 4.

$$\lim_{x \rightarrow 2} \frac{x^3 + 2x^2 - 1}{5 - 3x} = \frac{\lim_{x \rightarrow 2} (x^3 + 2x^2 - 1)}{\lim_{x \rightarrow 2} (5 - 3x)} = \frac{15}{-1} = -15.$$

Sometimes it is easy to compute the limits by Direct substitution.

Notice that

$$\lim_{x \rightarrow 2} x^3 + 2x^2 - 1 = 15 \quad \text{but if } f(x) = x^3 + 2x^2 - 1$$

$$f(2) = 2^3 + 2(2)^2 - 1 = 15.$$

$$\text{that is } \lim_{x \rightarrow 2} f(x) = f(2).$$

Direct Substitution Property. If f is a polynomial or a rational function and a is in the Domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Rmk. Functions that have the Direct Substitution property are called continuous at a .

Example. Find $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.

Soln. Let $f(x) = \frac{x^2 - 1}{x - 1}$. $\text{Dom } f = (-\infty, 1) \cup (1, \infty)$, we can't find the limit by substituting $x = 1$ b/c $f(1)$ is not defined. Now, we can't apply the Quotient Law b/c the limit of the denominator is 0. Instead, we need to do some preliminary algebra.

$$\text{if } x \neq 1 \text{ then } \frac{x^2 - 1}{(x - 1)} = \frac{(x - 1)(x + 1)}{(x - 1)} \quad \text{since } x \neq 1$$

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2.$$

Example. Evaluate $\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h}$.

Soln. Let $F(h) = \frac{(3+h)^2 - 9}{h}$ we know $(x+y)^2 = x^2 + 2xy + y^2$

$$= \frac{9 + 6h + h^2 - 9}{h} = \frac{6h + h^2}{h} = \frac{\cancel{h}(6+h)}{\cancel{h}} = \frac{6+h}{1}$$

that means

$$\lim_{h \rightarrow 0} F(h) = \lim_{h \rightarrow 0} 6+h = 6.$$

Example. Find $\lim_{t \rightarrow 0} \frac{\sqrt{t^2+9} - 3}{t^2}$.

Soln. We want apply $(a^2 - b^2) = (a-b)(a+b)$.

if you have $a+b \Rightarrow a-b$ is the conjugate

if you have $a-b \Rightarrow a+b$ is the conjugate

"the idea is rationalize the numerator"

$\underbrace{\sqrt{t^2+9}}_a - \underbrace{3}_b$ then $\sqrt{t^2+9} + 3$ is the conjugate.

then $(\sqrt{t^2+9} - 3)(\sqrt{t^2+9} + 3) = \cancel{(\sqrt{t^2+9})^2} - (3)^2 = t^2 + 9 - 9 = t^2$. (17)

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2+9} - 3}{t^2} = \lim_{t \rightarrow 0} \frac{\sqrt{t^2+9} - 3}{t^2} \times \frac{\sqrt{t^2+9} + 3}{\sqrt{t^2+9} + 3}$$

$$= \lim_{t \rightarrow 0} \frac{(\sqrt{t^2+9} - 3)(\sqrt{t^2+9} + 3)}{t^2(\sqrt{t^2+9} + 3)}$$

$$= \lim_{t \rightarrow 0} \frac{\cancel{t^2}}{\cancel{t^2}(\sqrt{t^2+9} + 3)} \quad \text{by (17).}$$

$$= \lim_{t \rightarrow 0} \frac{1}{\sqrt{t^2+9} + 3} = \frac{1}{\sqrt{0^2+9} + 3} = \frac{1}{3+3} = \frac{1}{6}$$

Using One-sided Limits.

Example. Show that $\lim_{x \rightarrow 0} |x| = 0$

Sln. We know that

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$\text{So, } \lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0.$$

$$\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = -\lim_{x \rightarrow 0} x = 0.$$

then by thm $\heartsuit$ $\lim_{x \rightarrow 0} |x| = 0$.

Example. If

$$f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4 \\ 8-2x & \text{if } x < 4. \end{cases}$$

determine whether $\lim_{x \rightarrow 4} f(x)$ exists.

Sln. $\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} (8-2x) = 8-2(4) = 0.$

$$\lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \sqrt{x-4} = 0$$

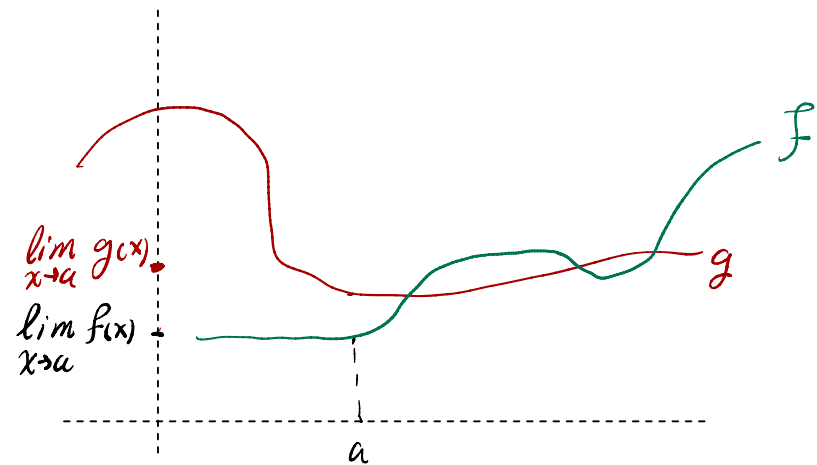
by thm $\heartsuit$ since right-left hand limits are equal then

$$\lim_{x \rightarrow 4} f(x) = 0.$$

Problem. Compute $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x}\right)$

The squeeze theorem. The following theorems describe how the limit of functions are related when the values of one function are greater than those of another.

Thm 1 If $f(x) \leq g(x)$ when x is near a and the limits of f and g exist as x approaches a , then.



$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x).$$